

Descriptive/Laboratory

1. A 0.01 M solution of a metal salt forms white precipitates when mixed with equal volumes of 0.01 M Na_2CO_3 and 0.01 M Na_2SO_4 , but no precipitate with 0.01 M NaOH. What is the metal salt?
- (A) MgCl_2 (B) ZnCl_2
(C) AgNO_3 (D) BaCl_2
2. A new balance is tested with replicate measurements of a standard 50.00 g mass, and gives readings of 49.31 g, 49.31 g, 49.29 g, and 49.30 g. Which description best characterizes the new balance?
- (A) Low accuracy and low precision
(B) Low accuracy and high precision
(C) High accuracy and low precision
(D) High accuracy and high precision
3. Oxalate ion, $\text{C}_2\text{O}_4^{2-}$, can be quantitated by titration with potassium permanganate in acidic solution. Which is used to identify the endpoint?
- (A) A persistent pink color from permanganate
(B) A persistent pink color from Mn^{2+}
(C) A persistent pink color from added phenolphthalein
(D) Bubbles of CO_2 are no longer observed
4. The number of moles of a weak monoprotic acid is determined by titrating with standardized NaOH solution to a phenolphthalein endpoint. The procedure calls for the acid to be dissolved in 50 mL of distilled water. What error would be introduced by using 75 mL of distilled water instead?
- (A) The number of moles would increase significantly.
(B) The number of moles would decrease significantly.
(C) The number of moles would not be significantly affected.
(D) The number of moles would be decreased by the change if the indicator concentration is low but increased if the indicator concentration is high.
5. Solutions of which metal salt give a yellow flame test?
- (A) Na (B) K (C) Cu (D) Ba
6. The concentration of which salt would be most difficult to measure using UV-visible spectroscopy?
- (A) NaMnO_4 (B) KClO_4
(C) $\text{Co}(\text{NO}_3)_2$ (D) CuCl_2
7. Which compound could be used to remove Ca^{2+} from hard water?
- (A) LiCl (B) Na_2CO_3
(C) KNO_3 (D) K_2SO_4
8. Addition of silver nitrate solution to 1.0 M solutions of which compounds will result in a yellow precipitate?
- I. NaCl II. KI
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
9. A student is cleaning a buret in preparation for a titration in which the buret will be filled with 0.100 M NaOH. What should be used for the final rinse of the buret?
- (A) Ethanol (B) Distilled water
(C) 0.100 M HCl (D) 0.100 M NaOH
10. In which solvent does solid iodine dissolve to give a purple solution?
- (A) Water (B) Ethanol
(C) Carbon tetrachloride (D) 0.1 M aqueous KI
11. Which glassware would be most suitable for preparing a 0.1500 M solution of silver nitrate?
- (A) 50 mL volumetric flask
(B) 50 mL round-bottom flask
(C) 50 mL Erlenmeyer flask
(D) 50 mL graduated cylinder
12. A 2.00 g sample of $\text{NaHCO}_3(s)$ is heated in a 10.00 g crucible to drive off H_2O and CO_2 . The mass data in the table were collected by heating the crucible and then periodically letting it cool to room temperature and reweighing it. Which is the first measurement after which heating could have been stopped?
- | Measurement | Mass of crucible and solid |
|-------------|----------------------------|
| 0 | 12.00 g |
| 1 | 11.48 g |
| 2 | 11.24 g |
| 3 | 11.25 g |
| 4 | 11.24 g |
- (A) Measurement 1 (B) Measurement 2
(C) Measurement 3 (D) Measurement 4

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13. Which apparatus would be most suitable for collecting a sample of solid barium sulfate produced through the reaction of aqueous solutions of barium hydroxide and sulfuric acid?
- (A) Barometer (B) Beaker
(C) Büchner funnel (D) Buret
14. A solution is prepared by dissolving 10.00 g of a salt in 90.00 g of water. What additional information is needed to calculate the molarity of this solution?
- I. The molar mass of the salt
II. The density of the solution
- (A) I alone would allow one to calculate the molarity.
(B) II alone would allow one to calculate the molarity.
(C) I and II together would allow one to calculate the molarity.
(D) The molarity cannot be calculated even if both I and II are known.
15. Which substance dissolves to give a colorless solution?
- (A) KClO_4 (B) K_2CrO_4
(C) KMnO_4 (D) $\text{K}_3\text{Fe}(\text{CN})_6$
16. Which substance does not give off heat when added to water?
- (A) NaOH (B) KNO_3
(C) H_2SO_4 (D) CaO
17. A solution of a nitrate salt does not form a precipitate when treated with dilute sulfuric acid. Which cation is not present in this solution?
- (A) NH_4^+ (B) Al^{3+}
(C) Fe^{2+} (D) Pb^{2+}
18. Which measurement would be least able to distinguish between a sample of elemental bromine prepared using exclusively the ^{79}Br isotope ($^{79}\text{Br}_2$) and one prepared using exclusively the ^{81}Br isotope ($^{81}\text{Br}_2$)?
- (A) Infrared spectroscopy (B) Liquid density
(C) Mass spectrometry (D) Vapor density
19. What is the main purpose of adding boiling chips to a solution that will be heated?
- (A) To reduce the boiling point of the solution
(B) To increase the boiling point of the solution
(C) To prevent the solution from superheating
(D) To decrease the time required for the solution to reach its boiling point
20. An aqueous solution contains one or more of the metal cations Mg^{2+} , Ba^{2+} , and Pb^{2+} at concentrations of 0.1 M. Addition of an equal volume of 0.1 M sodium sulfate causes formation of a precipitate. Which inference regarding the original solution is correct?
- (A) The solution must contain Mg^{2+} or Ba^{2+} .
(B) The solution must contain Mg^{2+} or Pb^{2+} .
(C) The solution must contain Ba^{2+} or Pb^{2+} .
(D) The solution must contain Mg^{2+} , Ba^{2+} , and Pb^{2+} .
21. Iron(III) nitrate is dissolved in water to form a 0.1 M Fe^{3+} solution and allowed to stand at room temperature for three days. During that time, a precipitate forms. Addition of which reagent will best prevent formation of a precipitate?
- (A) Sulfuric acid (B) Potassium hydroxide
(C) Sodium bicarbonate (D) Sodium chloride
22. Adding hydrogen peroxide to which reagent will NOT result in gas evolution?
- (A) Sodium hypochlorite (B) Copper(II) sulfate
(C) Manganese(IV) oxide (D) Sodium bisulfite
23. Acetylsalicylic acid is synthesized from salicylic acid, then isolated and dried. The melting point of the product is determined. If some salicylic acid is present, what will be observed? The melting point of pure acetylsalicylic acid is 136°C and the melting point of pure salicylic acid is 159°C .
- (A) The product will melt below 136°C .
(B) Most of the product will melt at 136°C , with some flecks of solid remaining until the temperature reaches 159°C , at which point the remaining material will melt.
(C) The product will melt between 136 and 159°C .
(D) The product will melt above 159°C .

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24. A 10.00 g sample of an unknown metal is reacted with hydrochloric acid to form the divalent metal ion and hydrogen gas, which is then collected over water. The molar mass of the metal is calculated to determine the identity of the metal. Which will cause the calculated molar mass to be too large?

I. Not correcting for the vapor pressure of water
II. Not allowing the reaction to go to completion

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

25. What is the mobile phase in thin layer chromatography?

- (A) A liquid solvent such as isopropanol
(B) A smooth solid surface such as a glass plate
(C) A coating on a solid surface such as silica gel
(D) A chamber such as a glass jar

26. Which is the strongest acid?

- (A) HBrO
(B) HBrO_2
(C) HBrO_3
(D) HBrO_4

27. Which compound, when dissolved in water, will increase the electrical conductivity of water the least?

- (A) SO_3 (B) H_2O_2
(C) MgCl_2 (D) NH_4Br

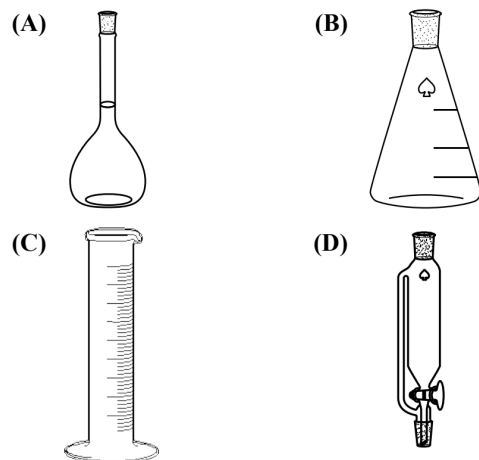
28. Which metal reacts most vigorously with water?

- (A) Aluminum (Al) (B) Zinc (Zn)
(C) Cesium (Cs) (D) Mercury (Hg)

29. A student determines the acetic acid concentration of a sample of distilled vinegar by measuring 25.00 mL of the vinegar with a volumetric pipet, then titrating the sample with standardized sodium hydroxide solution using phenolphthalein as an indicator. Which error will give an acetic acid content for the vinegar that is too high?

- (A) The titration flask has some distilled water in it when the acetic acid sample is added to it.
(B) The volumetric pipet has some distilled water in it when the acetic acid sample is measured with it.
(C) Some of the vinegar remains in the volumetric pipet rather than being dispensed into the titration flask.
(D) The endpoint is recorded when the solution turns dark red instead of faint pink.

30. Which 50-mL container would be most suitable for measuring and dispensing 37 mL of an aqueous solution?



31. Which salt gives a colorless aqueous solution?

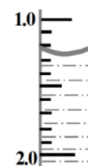
- (A) NaClO_4 (B) K_2CrO_4
(C) $\text{Ni}(\text{NO}_3)_2$ (D) CuSO_4

32. 100 mL of a 1.0 M solution of hydrochloric acid is gradually added to a stirred mixture containing 0.01 mol calcium hydroxide, 100 mL water, and a small amount of phenolphthalein. What changes are observed?

I. The mixture changes from colorless to pink.
II. The mixture changes from cloudy to clear.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

33. Below is shown a section of a buret filled with titrant before beginning a titration. What initial volume should be recorded?



- (A) 1.20 mL (B) 1.26 mL
(C) 2.74 mL (D) 2.80 mL

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34. Titration with aqueous triiodide solution to a starch endpoint would be LEAST suitable for which determination?
- (A) Analysis of nickel content in a Ni/Co alloy
(B) Determination of ascorbic acid in a vitamin C tablet
(C) Measurement of sulfite concentration in a white wine
(D) Determination of waters of hydration in a sodium thiosulfate sample
35. Which liquid will NOT react with metallic sodium?
- (A) Water (B) Mineral oil
(C) Ethanol (D) Carbon tetrachloride
36. The molar enthalpy of solution of a salt that dissolves endothermically in water is measured in a coffee-cup calorimeter by weighing a known amount of distilled water into the cup and measuring its temperature, then adding a known mass of the salt to the water and measuring the temperature after the salt dissolves. If some water is initially present in the cup before the weighed amount of water is added, what is the effect on the experiment?
- (A) Calculated $\Delta H^\circ_{\text{soln}} > \text{True } \Delta H^\circ_{\text{soln}}$
(B) Calculated $\Delta H^\circ_{\text{soln}} = \text{True } \Delta H^\circ_{\text{soln}}$
(C) Calculated $\Delta H^\circ_{\text{soln}} < \text{True } \Delta H^\circ_{\text{soln}}$
(D) The reaction will be calculated to be exothermic even though it is actually endothermic.
37. When substances are separated by fractional distillation, which property is least typical of the substance that distills first?
- (A) Highest molar mass
(B) Weakest intermolecular forces
(C) Greatest vapor pressure
(D) Lowest boiling point
38. Which element is most abundant (by mass) in the Earth's crust?
- (A) Carbon (B) Oxygen
(C) Magnesium (D) Silicon
39. Which gas has the highest molar solubility in water at 25 °C and 1 atm?
- (A) CO₂ (B) NH₃ (C) O₂ (D) H₂S
40. Which compound is colorless?
- (A) NaMnO₄ (B) CrAsO₄
(C) RbIO₄ (D) BaCrO₄
41. Chlorine gas is bubbled into a colorless aqueous solution of sodium iodide. Which is the best description of what takes place?
- (A) A precipitate of white NaCl forms.
(B) A precipitate of metallic Na forms.
(C) The solution turns pale green as the chlorine dissolves.
(D) The solution turns yellow-brown as iodide reacts with the chlorine.
42. A student determines the number of moles of water in a hydrated metal oxide by weighing a clean, dry crucible and lid while the crucible is empty, then reweighing the crucible and lid with a sample of the hydrate, heating the crucible and lid with a Bunsen burner and then reweighing the crucible and lid with the sample after cooling to room temperature. Which error will result in too high a value for the amount of water of hydration?
- (A) The heating is conducted only once instead of the three times recommended by the procedure.
(B) The lid is left off the crucible when it is weighed with the hydrated oxide.
(C) The metal oxide reacts partially with oxygen in the air, forming a compound in a higher oxidation state.
(D) Some of the heated oxide is spilled from the crucible before it can be weighed.
43. Which salt dissolves in water to give a pink solution?
- (A) Co(CH₃COO)₂ (B) CuSO₄
(C) Fe(NO₃)₃ (D) ZnCl₂
44. A student is using a buret for a titration. What initial buret reading should be recorded?
- 
- (A) 6.6 mL (B) 7.4 mL
(C) 6.63 mL (D) 7.37 mL

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45. A strip of metallic zinc is placed in a beaker containing dilute aqueous copper(II) nitrate. Which statement correctly describes what takes place?
- (A) No reaction takes place.
(B) The mass of the metal strip decreases as the zinc is oxidized.
(C) A white precipitate of CuNO_3 is formed.
(D) Bubbles of NO(g) form as the nitrate ion is reduced.
46. Which element does NOT have multiple allotropes?
- (A) Carbon (B) Oxygen
(C) Fluorine (D) Phosphorus
47. Which reaction is not exothermic?
- (A) Dilution of concentrated hydrochloric acid in water.
(B) Dilution of concentrated sulfuric acid in water.
(C) Dissolution of solid sodium hydroxide in water.
(D) Dissolution of solid sodium bicarbonate in water.
48. The molar mass of a volatile organic liquid is measured by weighing the mass of an empty flask of known volume, adding some of the liquid, heating the flask in a water bath until the liquid has just vaporized completely, then sealing the flask, letting it cool, and remeasuring the mass of the flask with the remaining organic compound. Which error will lead to a calculated value of the molar mass that is lower than the theoretical value?
- (A) The flask is not sealed promptly after the last of the compound vaporizes.
(B) The organic compound dimerizes to an appreciable extent in the gas phase.
(C) The barometric pressure was assumed to be 1 atm but in fact was greater than 1 atm.
(D) The volume of the flask used in the calculation is smaller than the true volume.
49. A solution of which compound gives a violet flame test?
- (A) LiBr (B) NaCl
(C) KNO_3 (D) B(OH)_3
50. Addition of 1 M HBr to 0.1 M solutions of which compound results in evolution of a colorless gas?
- (A) NaHSO_3 (B) NaHSO_4
(C) $\text{Ce(NH}_4)_2(\text{NO}_3)_6$ (D) KBrO_3
51. Which solid is least soluble in water at 298 K?
- (A) CaF_2 (B) AgF
(C) Ba(OH)_2 (D) CoSO_4
52. Which methods would be suitable for determining the concentration of an aqueous solution of KMnO_4 ?
- I. Visible spectrophotometry (colorimetry)
II. Redox titration
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
53. The concentration of sulfate ion in a solution is measured by precipitating the sulfate as BaSO_4 , filtering the precipitate on ashless filter paper, and heating the filter paper and precipitate in a tared crucible with a Bunsen burner. Which error will result in a sulfate concentration that is higher than the actual concentration?
- (A) The empty crucible contains a few drops of water when it is tared.
(B) A glass fiber filter is used instead of ashless filter paper.
(C) Some fine precipitate is not captured by the filter.
(D) Some of the sulfate-containing solution spills before the BaCl_2 solution is added.
54. Which is the best way to dispense liquids using a volumetric pipet?
- (A) The pipet is immersed in the liquid to be dispensed, then lifted with a gloved finger on the top and the liquid allowed to drain to the mark. The remaining contents are then allowed to drain into the desired container.
(B) The pipet is immersed in the liquid to be dispensed, then lifted with a gloved finger on the top and the liquid allowed to drain to the mark. The remaining contents are then allowed to drain into the desired container, with a pipet bulb used to gently blow out any residual droplets.
(C) The tip of the pipet is submerged below the surface of the liquid and suction is applied using a pipet bulb until the liquid rises above the level of the mark. A gloved finger is then applied to the top of the pipet and the liquid allowed to drain to the mark. The remaining contents are then allowed to drain into the desired container.
(D) The tip of the pipet is submerged below the surface of the liquid and suction is applied by mouth until the liquid rises to the level of the mark. The contents are then allowed to drain into the desired container.

Descriptive/Laboratory

55. Addition of 6 M HCl to which substance will NOT result in gas evolution?
(A) Al (B) Zn (C) K_2CO_3 (D) $NaNO_3$
56. A divalent metal ion dissolved in dilute hydrochloric acid forms a precipitate when H_2S is bubbled through the solution. Which ion is it?
(A) Ca^{2+} (B) Mn^{2+} (C) Zn^{2+} (D) Cd^{2+}
57. Which compound will form the most intensely colored 0.01 M aqueous solution?
(A) $KMnO_4$ (B) $KClO_4$
(C) $KAl(SO_4)_2$ (D) KI
58. A solution contains 0.1 M Sr^{2+} ions and 0.1 M Ag^+ ions. Addition of an equal volume of a 0.5 M solution of which reagent will cause precipitation of a strontium salt but not a silver salt?
(A) $NaNO_3$ (B) NaF
(C) NaOH (D) NaCl
59. Which would be most suitable for measuring 2.7 mL of ethanol for addition to a reaction with acidified dichromate?
(A) 10-mL graduated cylinder
(B) 10-mL volumetric flask
(C) 10-mL volumetric pipet
(D) 10-mL beaker
60. The molar mass of a solid carboxylic acid is determined by titrating a known mass of the acid with a standardized solution of NaOH to a phenolphthalein endpoint. Which errors will lead to a molar mass that is smaller than the actual molar mass?
I. Some of the acid is spilled when being transferred into the titration flask.
II. The endpoint is recorded when the solution is dark red in color rather than light pink.
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
61. A white ionic solid is dissolved in water. Addition of a solution of sodium chloride to this solution results in a white precipitate. What was the cation in the original ionic solid?
(A) Na^+ (B) Fe^{3+} (C) Ag^+ (D) Sr^{2+}
62. An element is a solid at room temperature but soft enough to be cut with an ordinary knife. When placed in water, the element reacts violently. What element is it?
(A) Na (B) Mg (C) Cu (D) Hg
63. A 0.1 M solution of which salt is the most basic?
(A) $NaNO_3$ (B) $NaClO_4$
(C) $NaHSO_4$ (D) $NaHCO_3$
64. Addition of small amounts of which solids to 4 M HCl will result in gas evolution?
I. Zn II. Na_2SO_3
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
65. A student determines the acetic acid concentration of a sample of distilled vinegar by titration of 25.00 mL of the vinegar with standardized sodium hydroxide solution using phenolphthalein as an indicator. Which error will give an acetic acid content for the vinegar that is too low?
(A) Some of the vinegar is spilled when being transferred from the volumetric flask to the titration flask.
(B) The NaOH solution is allowed to stand for a prolonged period after standardization and absorbs carbon dioxide from the air.
(C) The endpoint is recorded when the solution turns dark red instead of faint pink.
(D) The vinegar is diluted with distilled water in the titration flask before the NaOH solution is added.
66. Which piece of equipment would give the most precise delivery of 25.0 mL of a solution?
(A) 25-mL graduated cylinder
(B) 25-mL syringe
(C) 25-mL beaker
(D) 25-mL volumetric pipet
67. Dissolution of which salt in water results in a decrease in the temperature of the solution?
(A) $KHSO_4$ (B) NaOH
(C) $AlCl_3$ (D) NH_4NO_3

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68. What is observed when equal volumes of 0.1 M aqueous HCl and 0.01 M aqueous Na_2SO_3 are mixed?
- (A) Colorless solution and a white precipitate
(B) Colored solution and a white precipitate
(C) Colorless solution and a colored precipitate
(D) Colorless solution, no precipitate, and gas evolution
69. Which combination of dilute aqueous reagents will not produce a precipitate?
- (A) $\text{AgNO}_3 + \text{HCl}$ (B) $\text{NaOH} + \text{HClO}_4$
(C) $\text{BaBr}_2 + \text{Na}_2\text{SO}_4$ (D) $\text{ZnI}_2 + \text{KOH}$
70. A solution of a salt of which metal produces a bright red color in a flame test?
- (A) Lithium (B) Sodium
(C) Potassium (D) Copper
71. Which cation forms a colorless aqueous solution?
- (A) Co^{2+} (B) Ni^{2+} (C) Cu^{2+} (D) Zn^{2+}
72. Which 50-mL container would be most suitable for measuring and dispensing 37 mL of an aqueous solution?

